

6

MEASUREMENT AND SPACE

GEOMETRICAL FIGURES

Geometry is the study of points, angles, lines, shapes and solids. The word 'geometry' comes from the Greek *ge*, meaning earth or land, and *metra*, meaning 'a measure of'. It is used by artists, architects, engineers and physicists. It is useful in construction projects, computer imaging and video game design.



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Chapter outline

	Working mathematically				
6.01 Transformations	U	F		R	C
6.02 Composite transformations	U	F		R	C
6.03 Line symmetry	U	F		R	C
6.04 Rotational symmetry	U	F		R	C
6.05 Classifying triangles	U	F		R	C
6.06 Angle sum of a triangle	U	F	PS	R	C
6.07 Exterior angle of a triangle	U	F		R	C
6.08 Classifying quadrilaterals	U	F		R	C
6.09 Angle sum of a quadrilateral	U	F	PS	R	C
6.10 Properties of quadrilaterals	U	F		R	C

Wordbank

angle sum The total of the sizes of the angles in a shape, such as a triangle

axis of symmetry A line that divides a shape in half, each of which is the mirror image of the other (plural: axes)

exterior angle of a triangle An 'outside' angle of a triangle formed after extending one of the sides of the triangle

isosceles triangle A triangle with 2 equal sides

obtuse-angled triangle A triangle with one obtuse angle

quadrilateral A shape with 4 straight sides

translation The process of 'sliding' a shape a certain distance and direction

trapezium A quadrilateral with one pair of parallel sides

Quiz
Wordbank 6

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

6.01 Transformations
6.05 Classifying triangles • Properties of triangles
6.06 Angle sum of a triangle
6.07 Exterior angle of a triangle 1 • Exterior angle of a triangle 2
6.10 Classifying quadrilaterals • Properties of quadrilaterals

Twig video (1):

6.01 Transformations: Skateboarding

Quizzes (5):

- Wordbank 6
- SkillCheck 6
- Mental skills 6
- Language of maths 6
- Test yourself 6

Skillsheets (2):

SkillCheck Naming shapes
6.03, 6.04 Line and rotational symmetry
6.08 Naming shapes

Worksheets (13):

6.01, 6.02, 6.03 5 mm grid paper
6.01 Transformations
6.03 Symmetry
6.04 Symmetry • Symmetry of flat shapes
6.05 Properties of triangles • Constructing triangles and quadrilaterals
6.07 Triangle geometry
6.08 Classifying quadrilaterals • Constructing triangles and quadrilaterals
6.10 Properties of quadrilaterals • Diagonal properties of quadrilaterals • Always, sometimes, never true
6.10 Equal angles
Mind map Geometrical figures

Puzzles (11):

6.05 Classifying triangles
6.07 Angles in triangles
6.08 Constructions group clues • What shape am I? • Singing in the car
6.09 Find the unknown angle • Mixed angle problems • Triangles and quadrilaterals
6.10 Geometry crossword • Shapes puzzle 1
Language of maths Geometry find-a-word

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- translate, reflect and rotate geometrical figures, and composite transformations of these
- identify line and rotational symmetries of geometrical figures
- classify triangles by sides and by angles
- identify convex and non-convex quadrilaterals
- solve problems involving the angle sum of a triangle and the exterior angle of a triangle
- identify and classify the special quadrilaterals and their properties: square, rectangle, parallelogram, rhombus, kite and trapezium
- solve problems involving the angle sum of a quadrilateral

Skillsheet
Naming shapes

Quiz
SkillCheck 6

SkillCheck ANSWERS ON P. 578

1 Draw:

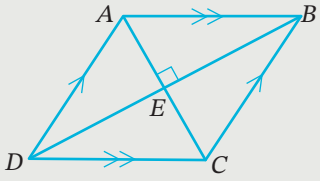
a 2 perpendicular lines

b 2 parallel lines.

2 Draw a rectangle and all its diagonals.

9780170477963

3 For this **quadrilateral**:



- **a** name 2 intervals that are parallel
- **b** are the diagonals equal in length?
- **c** what is the size of $\angle DEC$?
- **d** if $\angle DAB$ and $\angle ABC$ are cointerior and $\angle DAB = 115°$, what is the size of $\angle ABC$?

4 Draw a pair of parallel lines crossed by a transversal and mark a pair of alternate angles.

5 **a** Draw a parallelogram and label it *DEFG*.

- **b** Mark both pairs of parallel sides.
- **c** Name both pairs of parallel sides.
- **d** Mark the equal sides *DG* and *EF*

6 Draw a **triangle** that has:

- **a** a right angle
- **b** an obtuse angle.

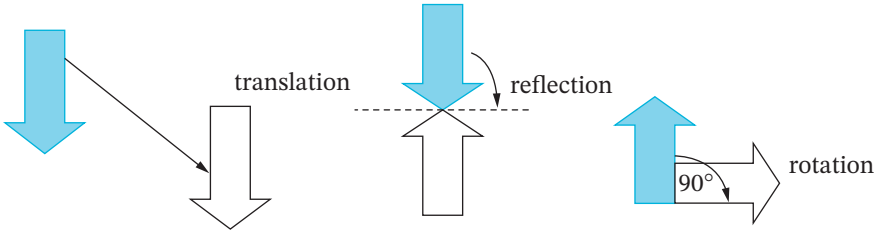
7 Draw 2 parallel lines crossed by a transversal and mark a pair of corresponding angles.

Transformations 6.01

Patterns in tiles, wallpaper and paving are usually made by taking a basic shape and repeating it. A pattern can be created by **sliding**, **flipping** or **spinning** the shape.

These movements have special names.

- A 'slide' is called a **translation**
- A 'flip' is called a **reflection**
- A 'spin' is called a **rotation**



Video
Transformations: Skateboarding

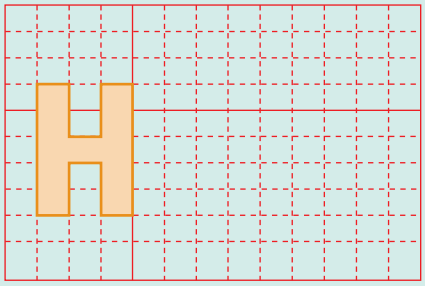
These movements are called **congruence transformations**, where 'congruence' means identical and 'transformation' means change. Even though a shape has changed position (transformed), it still has the same shape and size (congruence).

- The original shape is called the **original**
- The transformed shape is called the **image**

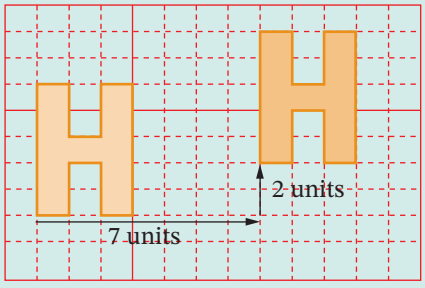
Video
Transformations

Example 1

Translate the H-shape below 7 units right and 2 units up.



Solution

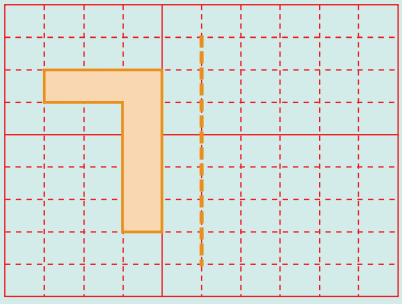


Every point on the original is shifted 7 units right and 2 units up to create the image.

Video
Transformations

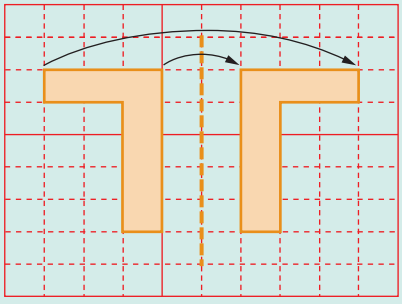
Example 2

Reflect the L-shape across the dotted line.



The dotted line is called the **line of reflection**.

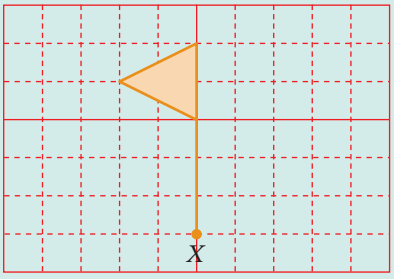
Solution



The original and its mirror-image are the same distance from the line of reflection.

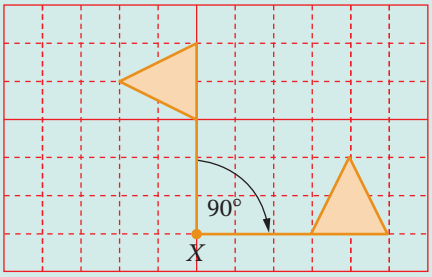
Example 3

Rotate the flag shape 90° in a clockwise direction about the point X.



X is called the **centre of rotation**.

Solution



The original and image are the same distance from the centre of rotation, so point X does not move.

Video
Transformations

6.01

EXERCISE 6.01 ANSWERS ON P. 578

Transformations U F R C

Worksheet
Transformations

Questions **1** to **4** of this exercise can also found on a worksheet for students to write in.

1 Translation, reflection or rotation? C

a To make a shape back-to-front to create a mirror-image

b To slide or shift a shape

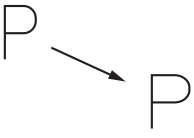
c To flip a shape

d To turn a shape upside-down

e To move a shape up, down or diagonally

f To spin a shape

g



h



i



j



k



l



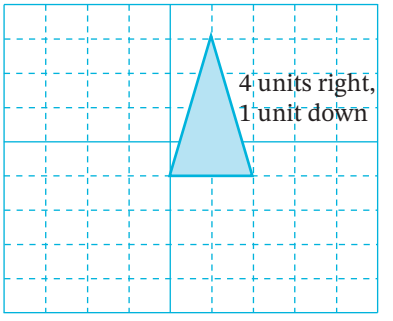
Foundation Standard Complex

EXAMPLE 1

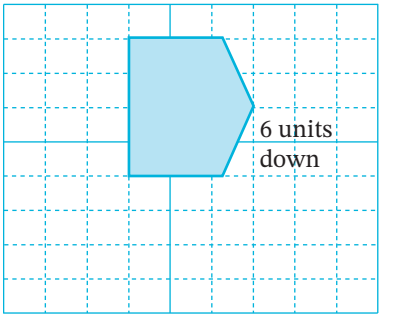
Worksheet
5 mm grid paper

2 Copy each shape onto grid paper and leave enough space to translate it according to the directions given.

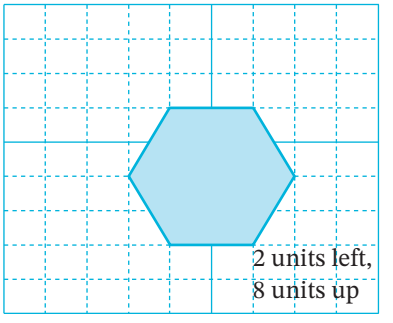
a



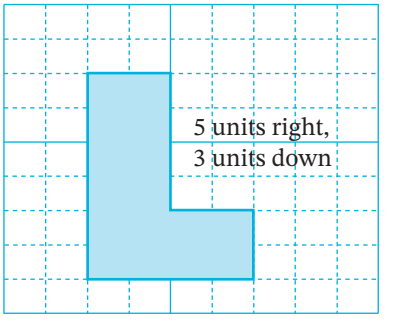
b



c



d

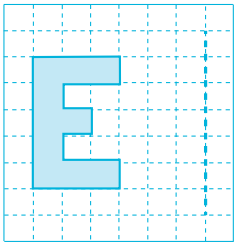


EXAMPLE 2

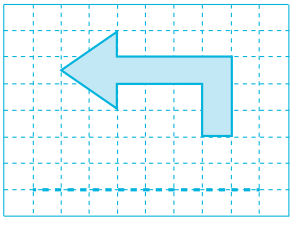
Worksheet
5 mm grid paper

3 Copy each shape onto grid paper and leave enough space to reflect it across the dotted line. Matching points on the original and reflected shape should be the same distance from the line of reflection.

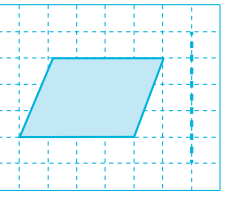
a



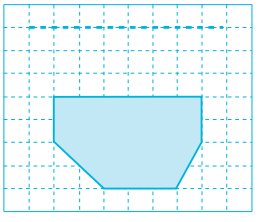
b



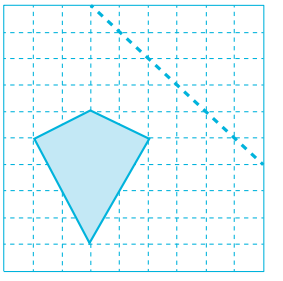
c



d



e



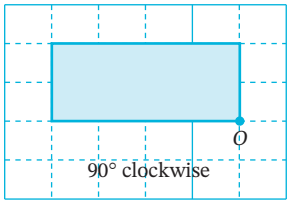
Foundation Standard Complex

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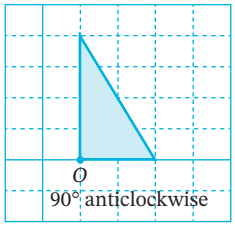
4 Copy each shape onto grid paper and leave enough space to rotate it by the given angle about the point O.

EXAMPLE 3

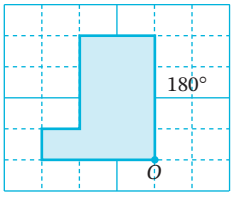
a



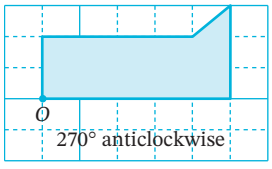
b



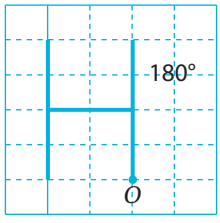
c



d



e



5 Why don't rotations of 180° need the label 'clockwise' or 'anti-clockwise'? C

6 What clockwise rotation gives the same result as: R

a a 90° rotation anti-clockwise?

b a 270° rotation anti-clockwise?

c a 90° rotation clockwise 3 times?

d a 90° rotation anti-clockwise followed by a 270° rotation clockwise?

7 What single translation gives the same result as a translation of: R C

a 3 units right, 1 unit up followed by 4 units left, 4 units up?

b 2 units left, 8 units down followed by 2 units right, 5 units up?

8 Which single translation gives the same result as 4 units right, 2 units up, 5 units right, 4 units down? Select the correct answer **A**, **B**, **C** or **D**. R C

A 9 right, 2 down

B 11 right, 2 down

C 6 right, 2 up

D 9 right, 2 up

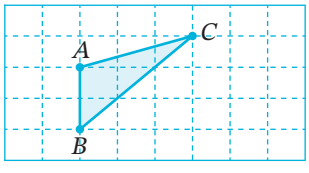
Technology

Transformations

Use dynamic geometry software to explore transformations.

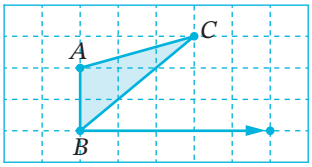
Translations

1 Draw and label a triangle $\triangle ACB$.

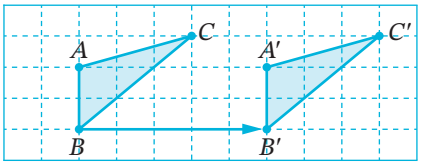


Foundation Standard Complex

2 Click point B and draw a vector by clicking a point 5 units to the right as shown.

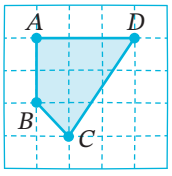


3 Use the vector to translate the entire triangle.



When an object or image is moved, the copy that is made is represented by letters with 'dashes' after each letter. You should now see triangle $A'B'C'$ on your screen. It is an exact copy of the original triangle ABC and has been translated 5 units to the right of the original triangle ABC.

4 Now draw, label and translate quadrilateral $ABCD$ 3 units down and 4 units to the left of point A.



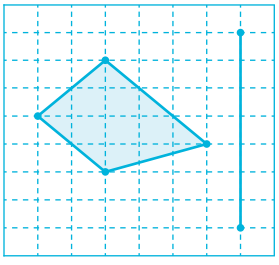
Reflections

1 Draw this quadrilateral and interval.

2 Use the software to reflect the quadrilateral in the line.

3 Drag any vertex of the original shape to see its reflection change.

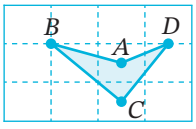
4 Draw some other shapes and reflect each one.



Rotations

1 Draw and label this quadrilateral.

2 Click on the shape and rotate it around point D 180° clockwise.



3 Now rotate the shape around the point using different angles and choose clockwise or anticlockwise.

4 Now rotate the shape around a different point using different angles.

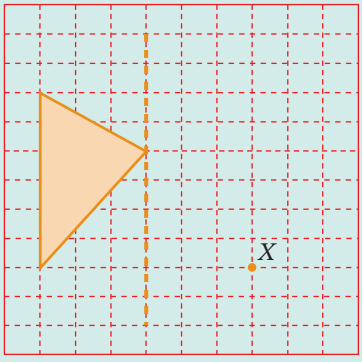
5 Now experiment by creating your own shapes and rotating them around a point using different angles.

Composite transformations 6.02

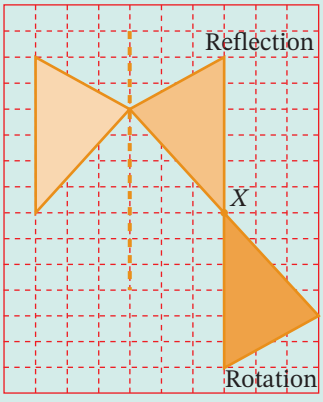
Combinations of translations, reflections and rotations can be applied one after the other. These are called **composite transformations**.

Example 4

Reflect the triangle below across the dotted line and then rotate the image 180° about X.



Solution

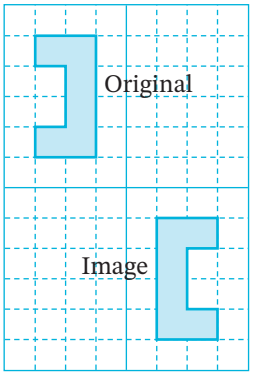


EXERCISE 6.02 ANSWERS ON P. 579

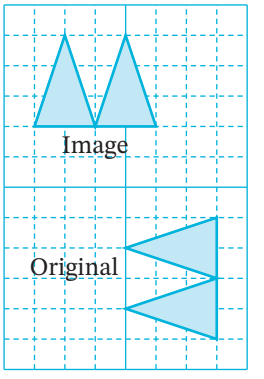
Composite transformations U F R C

1 For each diagram below, state the combination of 2 transformations used on the original figure to form the image. R C

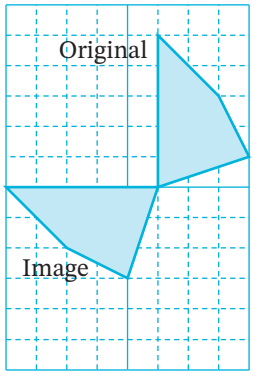
a



b



c

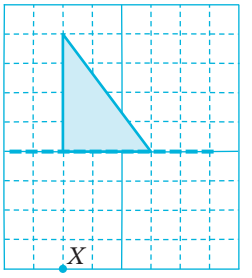


EXAMPLE 4

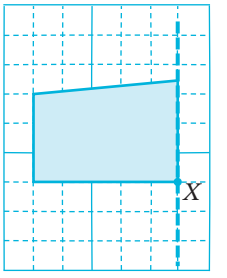
2 Copy each shape on grid paper and leave enough space to draw the final image when it is reflected in the given line and then rotated 90° clockwise about the point X.

Worksheet
5 mm grid paper

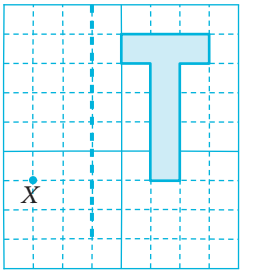
a



b

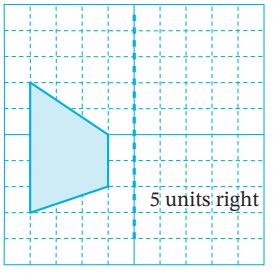


c

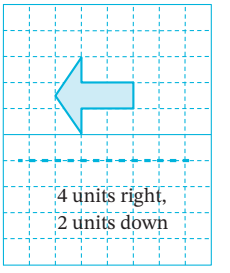


3 Copy each shape on grid paper and leave enough space to draw the final image when it is reflected in the given line and then translated by the given distance and direction.

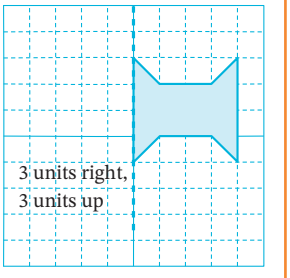
a



b



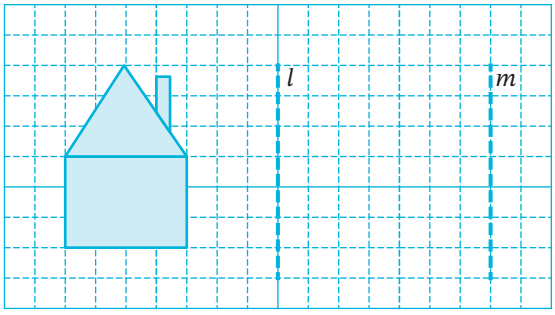
c



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9780170477963

4 **a** Copy the house shape on grid paper and leave enough space to draw the final image when it is reflected across line l and then across line m.



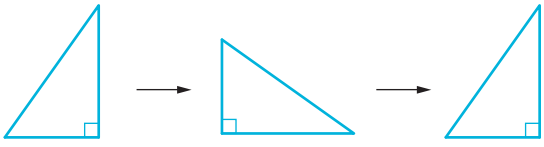
b Which single transformation would give the same result as the 2 reflections in part **a**? R C

5 Each diagram below has been transformed twice. R C

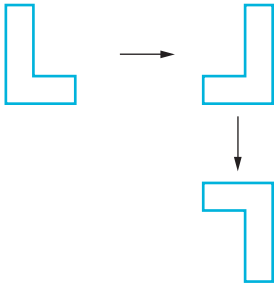
i Name the 2 transformations that have been performed.

ii Name one transformation that would give the same result as the 2 transformations.

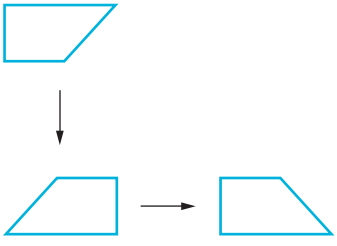
a



b



c



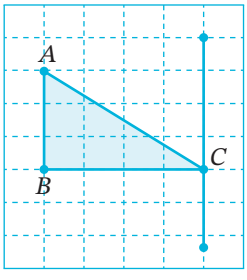
Technology

Composite transformations

This activity will use dynamic geometry software to explore composite transformations.

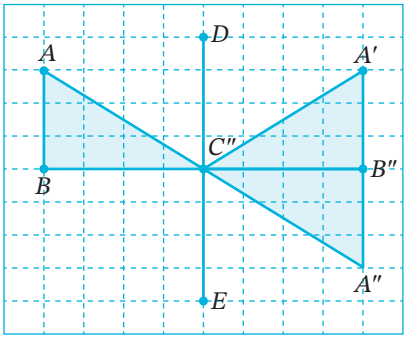
Triangle transformation

1 Draw and label the triangle shown. Draw the interval as shown as a line of reflection. Use the software to reflect the triangle across the line.



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2 Now reflect the reflected triangle A′B′C′ across its base B′C′.

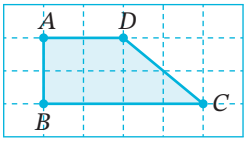


3 There is another way to do this combined transformation. Draw the triangle shown in step **1** again. Rotate the triangle about point C 180° clockwise.

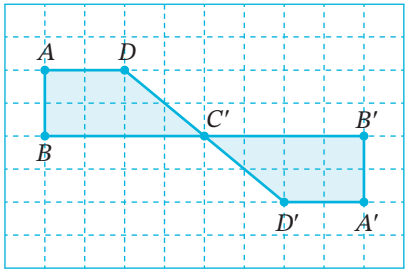
4 Compare this with the method used in steps **2** and **3** above. What do you notice?

Trapezium transformation

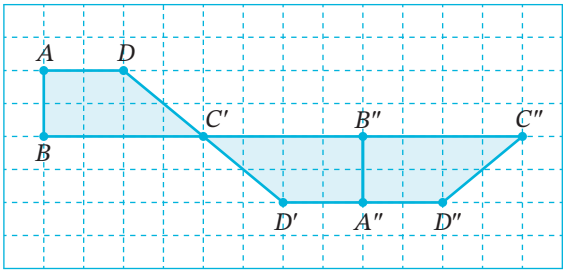
1 Draw and label this trapezium.



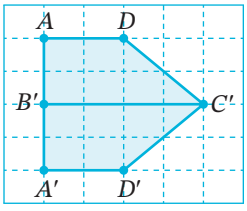
2 Rotate the trapezium clockwise, 180°.



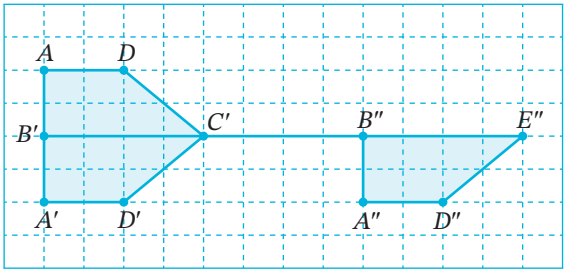
3 Now reflect the rotated trapezium A′B′C′D′ across the line A′B′.



4 Again, there is another way to do the same composite transformation. Draw the trapezium shown in step **1** again and reflect it across its base BC.



5 Translate the reflected trapezium 8 units to the right.



6 Compare this method of reflection and translation with the rotation and reflection used in steps **2** and **3**. What do you notice?

7 Now create your own composite transformation, using any combination of translation, reflection and rotation.

8 Swap your composite transformation with your group members and try to find an alternative method (or more if possible) to the composite transformation that they have created.

Line symmetry

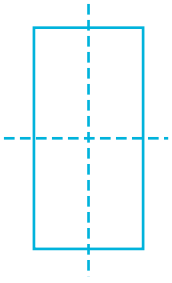
6.03

Symmetrical objects appear balanced and are pleasing to look at. The 2 main types of symmetry are **line symmetry** and **rotational symmetry**.

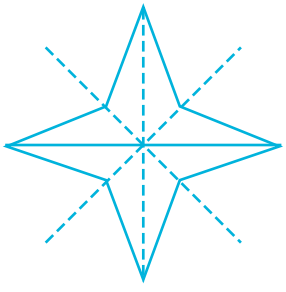
The 2 shapes below have line symmetry. If you fold along the line, one half of the shape will fit exactly on top of the other half. One half is the reflection or mirror-image of the other half. The fold line is called the **axis of symmetry**. The plural of 'axis' is **axes**.

Skillsheet
Line and rotational symmetry

Worksheet
Symmetry



This rectangle has 2 axes of symmetry.



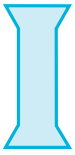
This star shape has 4 axes of symmetry.

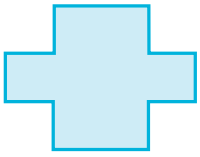
EXERCISE 6.03 ANSWERS ON P. 580

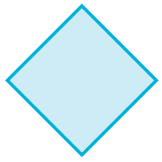
Line symmetry U F R C

1 Copy each shape and mark in its axes of symmetry. R

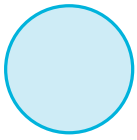
a 

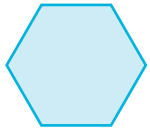
b 

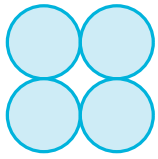
c 

d 

e 

f 

g 

h 

2 Which shapes in question **1** have: R

a one axis of symmetry?

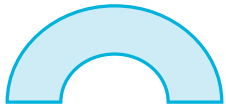
b 4 axes of symmetry?

c an infinite number of axes of symmetry?

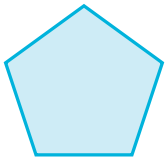
d 2 axes of symmetry?

3 Count the number of axes of symmetry of each shape. R

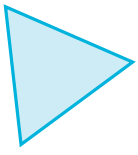
a 

b 

c 

d 

e 

f 

4 Copy the capital letters below that have line symmetry and draw their axes of symmetry.

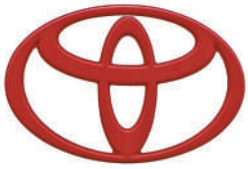
D H I K M N O R S V X Z

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5 How many axes of symmetry has each car logo? R

a 

b 

c 

d 

e 

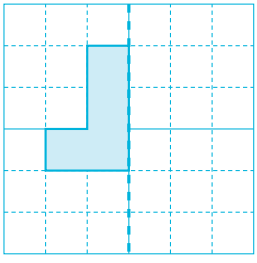
f 

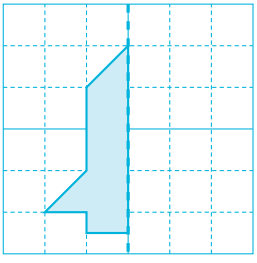
Left to right: Alamy Stock Photo/Wdnet Studio, Alamy Stock Photo/James Houck, Alamy Stock Photo/Peter Probst

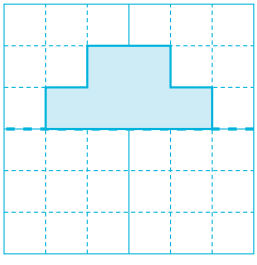
Left to right: Shutterstock.com/tanuha2001, Shutterstock.com/Martin Good, Alamy Stock Photo/Peter Probst

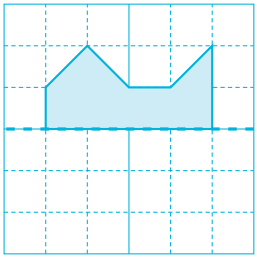
6 Use the Internet to find an example of a logo that has line symmetry.

7 Copy each figure onto grid paper and complete it so that the dotted line is an axis of symmetry.

a 

b 

c 

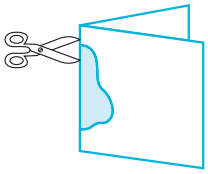
d 

6.03

Worksheet
5 mm grid paper

Foundation | Standard | Complex

8 Fold a piece of paper in half and draw a figure on one side of the folded edge, as shown in the diagram. Cut your shape out. What do you notice? C



9 Fold a piece of paper in half and place a few drops of paint or ink in the fold. What pattern do you get when you press the sides together? R C

10 Draw a shape that has: R

a no axes of symmetry

b 1 axis of symmetry

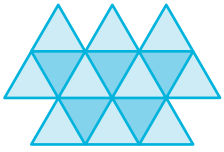
c 2 axes of symmetry

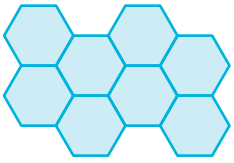
d 3 axes of symmetry

Did you know?

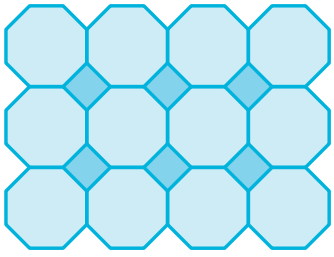
Tessellations

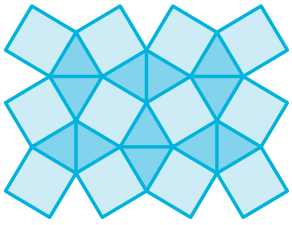
A tessellation is a pattern of shapes that fit together perfectly, leaving no gaps between the shapes. It is like tiling an infinite floor with shapes so that there are no gaps or overlaps.



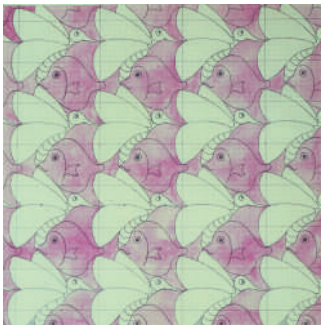


Sometimes it is necessary to use more than one shape to make sure that there are no gaps.





Maurits Cornelis Escher (1898–1972) was a Dutch graphic artist who created some interesting tessellations, such as this one.



M.C. Escher's "Symmetry Drawing E27"
© 2018 The M.C. Escher Company-The Netherlands. All rights reserved.

Which of these shapes tessellate?

kite	**oval**	**parallelogram**	**rhombus**
hexagon	**rectangle**	**octagon**	**pentagon**

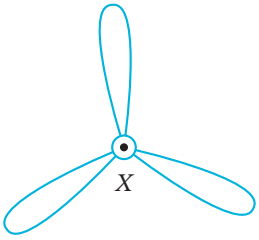
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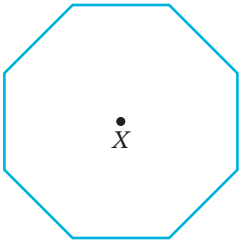
Rotational symmetry

6.04

The 2 shapes below have **rotational symmetry**. If you rotate each shape about the point X, it will fit exactly on top of itself at least once before one full revolution (360°). The centre point, X, is called the **centre of symmetry**. The number of times that a shape fits on itself in one revolution is called its **order of rotational symmetry**.



This propeller fits on itself 3 times in a full turn, so it has rotational symmetry **of order 3**.



This regular octagon fits on itself 8 times in a full turn, so it has rotational symmetry **of order 8**.

Skillsheet
Line and rotational symmetry

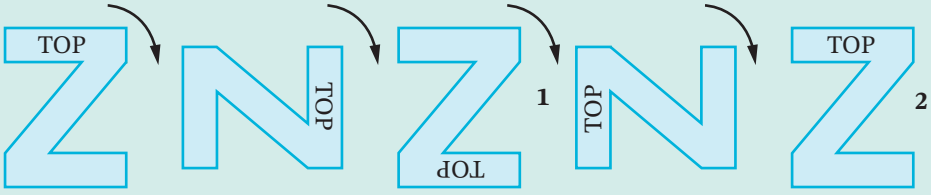
6.04

Worksheets
Symmetry
Symmetry of flat shapes

Example 5

What order of rotational symmetry does the letter Z have?

Solution



Z has rotational symmetry of order 2.

EXERCISE 6.04 ANSWERS ON P. 580

Rotational symmetry U F R C

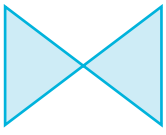
EXAMPLE 5

1 For each shape, state whether or not it has rotational symmetry, and if it does, copy the shape, mark its centre of symmetry and find its order of rotational symmetry. R C

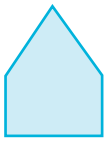
a



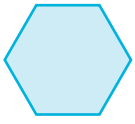
b



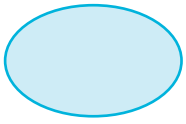
c



d



e



f



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2 Copy the capital letters below that have rotational symmetry and mark their centre of symmetry. R

D H I K M N O R S V X Z

3 For the car logos in Exercise 6.03 on page 247, state which ones have rotational symmetry and their order of rotational symmetry. R C

4 Use the Internet to find an example of a logo that has rotational symmetry.

5 For each shape: R C

i find how many axes of symmetry it has

ii state whether it has rotational symmetry and if it does, state the order

a



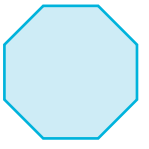
b



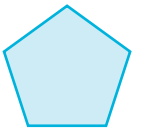
c



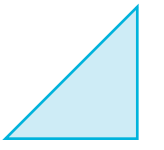
d



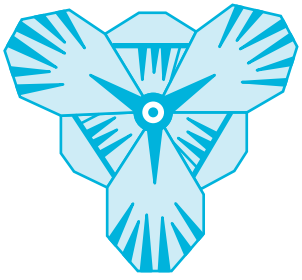
e



f



6 What order of rotational symmetry does this shape have, what is the angle of rotation? Select the correct answer **A**, **B**, **C** or **D**. R



A 6, 60°

B 3, 60°

C 2, 180°

D 3, 120°

7 Draw a shape that has: R

a no rotational symmetry

b rotational symmetry of order 2

c rotational symmetry of order 3

d rotational symmetry of order 4

8 What shape has an *infinite* order of rotational symmetry? R

Mental skills 6: Maths without calculators ANSWERS ON P. 580

Quiz
Mental skills 6

Estimating answers

A quick way of estimating an answer is to round each number in the calculation.

1 Study each example.

a $631 + 280 + 51 + 43 + 96 \approx 600 + 300 + 50 + 40 + 100$
$= (600 + 300 + 100) + (50 + 40)$
$= 1000 + 90$
$= 1090$ (Actual answer = 1101)

b $55 + 132 - 34 + 17 - 78 \approx 60 + 130 - 30 + 20 - 80$
$\approx (60 + 20 - 80) + (130 - 30)$
$= 0 + 100$
≈ 100 (Actual answer = 92)

c $67 \times 13 \approx 70 \times 12$
$= 840$ (Actual answer = 871)

d $78 \times 7 \approx 80 \times 7$
$= 560$ (Actual answer = 546)

e $929 \div 5 \approx 1000 \div 5$
$= 200$ (Actual answer = 185.8)

f $510 \div 24 \approx 500 \div 20$
$= 50 \div 2$
$= 25$ (Actual answer = 21.25)

2 Now estimate each answer.

a $27 + 11 + 87 + 142 + 64$

b $55 + 34 - 22 - 46 + 136$

c $684 + 903$

d $35 + 81 + 110 + 22 + 7$

e $517 - 96$

f $210 - 38 - 71 + 151 - 49$

g $766 - 353$

h 367×2

i 83×81

j 984×16

k $828 \div 3$

l $507 \div 7$

6.05 Classifying triangles

Worksheets
Properties of triangles
Constructing triangles and quadrilaterals

Puzzle
Classifying triangles

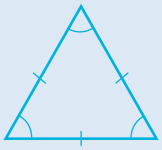
Video
Properties of triangles

Triangles can be classified in 2 ways:

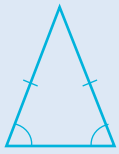
- by their **sides**: **equilateral**, **isosceles** or **scalene**
- by their **angles**: **acute-angled**, **obtuse-angled**, **right-angled**

Types of triangles

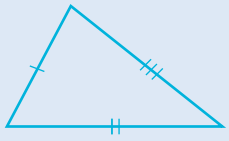
Sides



Equilateral
3 equal sides

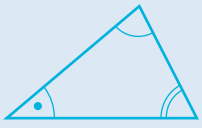


Isosceles
2 equal sides

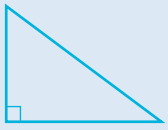


Scalene
No equal sides

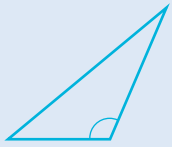
Angles



Acute-angled
3 acute angles



Right-angled
1 right angle

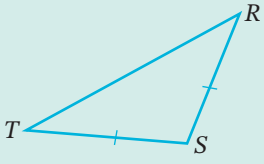


Obtuse-angled
1 obtuse angle

Video
Classifying triangles

Example 6

Classify this triangle by sides and angles.



Solution

The triangle has 2 equal sides so it is **isosceles**.

The triangle has an obtuse angle, $\angle S$, so it is **obtuse-angled**.

The triangle is **isosceles** and **obtuse-angled**.

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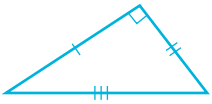
Classifying triangles U F R C

EXAMPLE 6

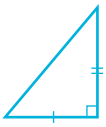
6.05

1 Classify each triangle by sides and by angles. R C

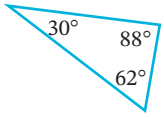
a



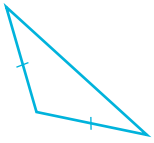
b



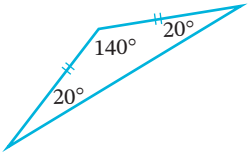
c



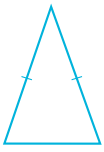
d



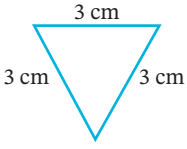
e



f



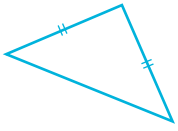
g



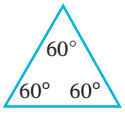
h



i



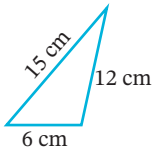
j



k



l



2 Draw an example of: R C

a a scalene triangle
b a right-angled scalene triangle
c an isosceles triangle
d an equilateral triangle
e a right-angled isosceles triangle
f an acute scalene triangle

3 Copy each triangle and draw its axes of symmetry. R

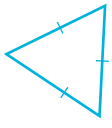
a



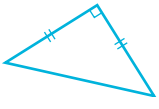
b



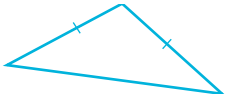
c



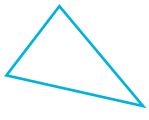
d



e



f



4 Do any triangles have rotational symmetry? Give examples to support your answer. R C

5 Is it possible to draw an equilateral right-angled triangle? Why? R C

6 The prefix 'tri' means '3'. Find the meaning of these mathematical 'tri' words. C

a trisect **b** trilateral **c** triangulate

7 Is it possible to draw a triangle with 2 obtuse angles? Why? R C

8 Which triangle is both obtuse-angled and scalene? Select **A**, **B**, **C** or **D**. C

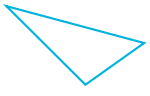
A



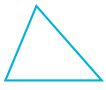
B



C



D



Foundation Standard Complex

9 Draw and cut out an isosceles triangle. R C

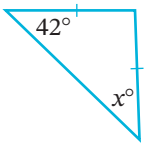
- **a** How many axes of symmetry does an isosceles triangle have?
- **b** By folding, mark the equal angles in an isosceles triangle.
- **c** Describe in words where the equal angles in an isosceles triangle are.

10 Draw and cut out an equilateral triangle. R C

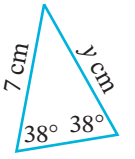
- **a** How many axes of symmetry does an equilateral triangle have?
- **b** By folding, mark the equal angles in an equilateral triangle.
- **c** Describe in words where the equal angles in an equilateral triangle are.

11 Find the value of the variable(s) in each diagram, giving a reason. R C

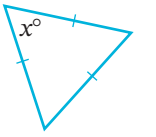
a



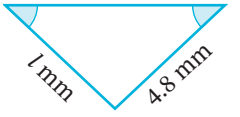
b



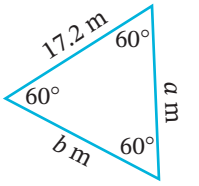
c



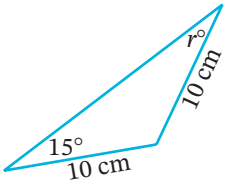
d



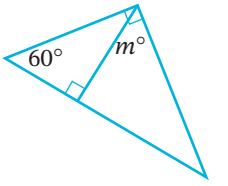
e



f



12 Find m. Select **A**, **B**, **C** or **D**. R



A 45° **B** 60°

C 30° **D** 120°

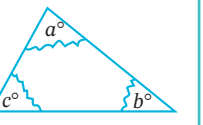
Investigation

Angle sum of a triangle

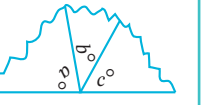
What is the sum of the 3 angles in a triangle? In groups of 2 to 4, complete this activity.

Paper-cutting activity

1 Draw a large triangle on paper, cut it out and label its 3 angles a, b and c.



2 Tear off the 3 angles and arrange them next to each other, so that their points meet.



3 What type of angle do they form? How many degrees are in this type of angle?

4 To see if this works for all triangles, repeat the above steps for different triangles.

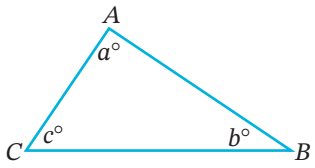
Foundation Standard Complex

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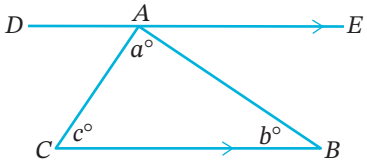
Formal proof

We can use parallel lines to prove that the angle sum of a triangle is 180°.

1 Draw any triangle ABC, with angles of size $a°$, $b°$ and $c°$.



2 Draw a line DE parallel to CB through A.



3 Which angle in $\triangle ABC$ is equal to $\angle DAC$? Why?

4 Which angle in $\triangle ABC$ is equal to $\angle EAB$? Why?

5 What is $\angle DAC + a° + \angle EAB$? Why?

6 What does this show about $a° + b° + c°$?

7 What does this show about the angle sum of a triangle?

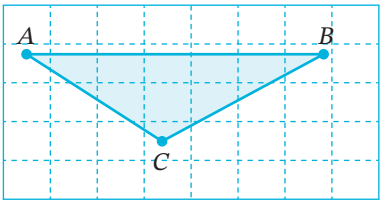
8 Check by measuring with a protractor.

Technology

Angle sum of a triangle

Use dynamic geometry software to find the angle sum of a triangle.

1 Draw and label any triangle ABC.



2 Measure the size of each angle.

3 Find the angle sum of the triangle.

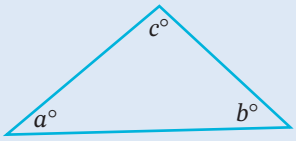
4 Move each vertex in the triangle that you have drawn and watch the size of each angle change. What is the angle sum of any triangle?

6.06 Angle sum of a triangle

Angle sum of a triangle

The **angle sum** of a triangle is 180°

$a + b + c = 180$

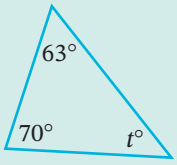


Video
Angle sum of a triangle

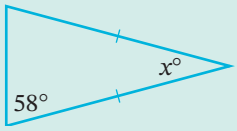
Example 7

Find the value of each variable, giving reasons.

a



b



Solution

a $t + 70 + 63 = 180$ (angle sum of a triangle)

$t + 133 = 180$

$t = 180 - 133$

$= 47$

b $x + 58 + 58 = 180$ (angle sum of an isosceles triangle)

$x + 116 = 180$

$x = 180 - 116$

$= 64$

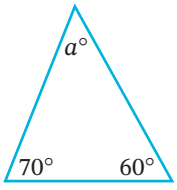
EXERCISE 6.06 ANSWERS ON P. 580

EXAMPLE 7

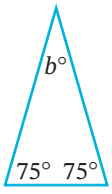
Angle sum of a triangle U F PS R C

1 Find the value of each variable. R C

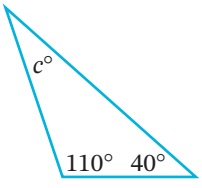
a



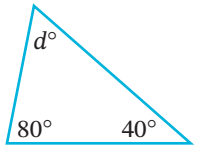
b



c



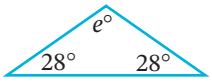
d



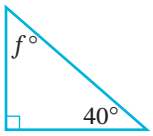
Foundation Standard Complex

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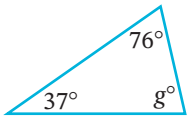
e



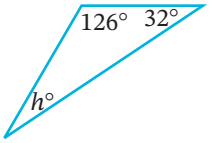
f



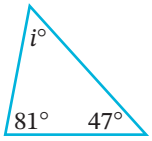
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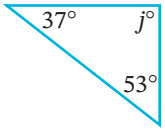
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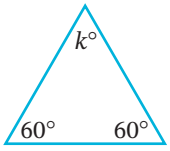
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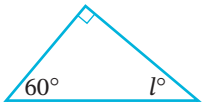
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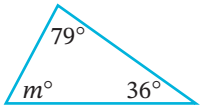
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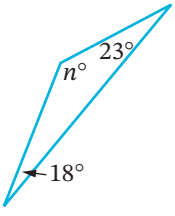
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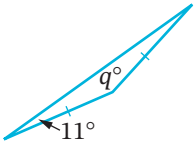
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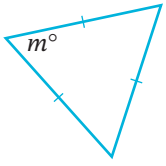
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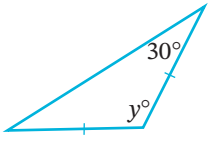
o



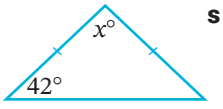
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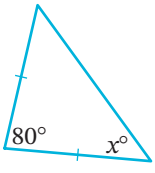
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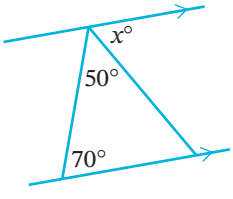


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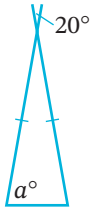


2 Using what you know about angles, find the value of each variable, giving reasons.
PS R C

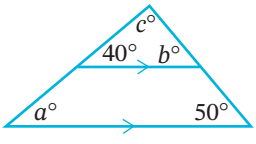
a



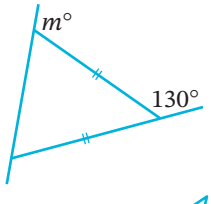
b



c

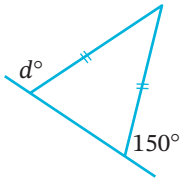


3 What is the value of m in this diagram?
Select the correct answer **A**, **B**, **C** or **D**. PS R



A 50 **B** 100
C 115 **D** 130

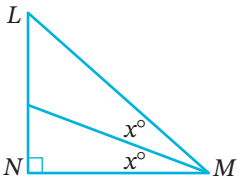
4 What is the value of d in this diagram? Select **A**, **B**, **C** or **D**. PS R



A 30 **B** 120
C 150 **D** 60

Foundation Standard Complex

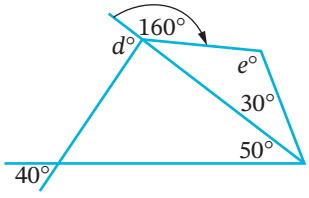
5 *LMN* is an isosceles right-angled triangle. What is the value of x? Select **A**, **B**, **C** or **D**. PS R



A 22.5 **B** 30

C 45 **D** 60

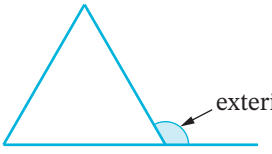
6 In the diagram, find the value of d and e, giving reasons. PS R C

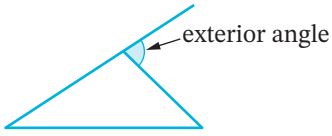


Investigation

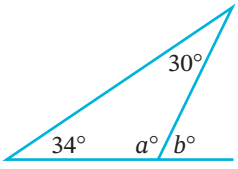
Exterior angle of a triangle

An **exterior angle** of a triangle is created by extending one side of the triangle. 'Exterior' means 'outside', while 'interior' means 'inside'.





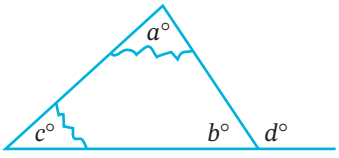
What is the relationship between the exterior angle of a triangle and the interior angles?



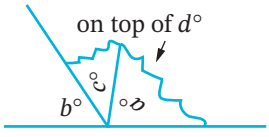
1. Find the value of a and b in this triangle.
2. Which is the exterior angle: a or b?
3. How is the exterior angle related to **2** of the (interior) angles of this triangle?
4. Copy and complete: An exterior angle of a triangle is equal to the _____ of the interior opposite _____.

Paper-cutting activity

1. Draw a large triangle on paper and label its 3 angles $a°$, $b°$ and $c°$.
2. Extend one of the sides of the triangle and label the exterior angle $d°$. Cut out the triangle as well as angle $d°$.



3. Tear off the 2 angles $a°$ and $c°$ that are not adjacent to $d°$ and arrange them next to each other on top of angle $d°$.
4. Do the 2 interior angles fit exactly on top of the exterior angle?
5. Check by measuring with a protractor.
6. To see if this works for all triangles, repeat the above steps for different triangles.



Foundation Standard Complex

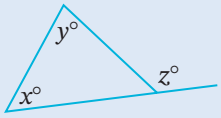
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Exterior angle of a triangle 6.07

Exterior angle of a triangle

An exterior angle of a triangle is equal to the sum of the 2 interior opposite angles.

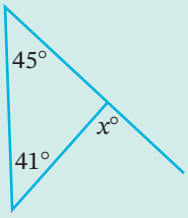
$z = x + y$



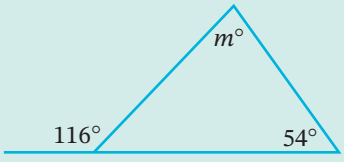
Example 8

Find the value of each variable, giving reasons.

a



b



Solution

a $x = 45 + 41$ (exterior angle of a triangle)

$= 86$

b $m + 54 = 116$ (exterior angle of a triangle)

$m = 116 - 54$

$= 62$

EXERCISE 6.07 ANSWERS ON P. 581

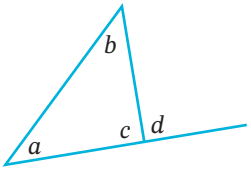
Exterior angle of a triangle U F R C

1 For each triangle below, name: R C

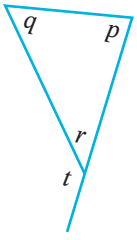
i the exterior angle

ii the 2 interior angles opposite the exterior angle.

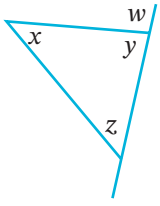
a



b



c



Worksheet
Triangle geometry

Puzzle
Angles in triangles

Videos
Exterior angle of a triangle 1
Exterior angle of a triangle 2

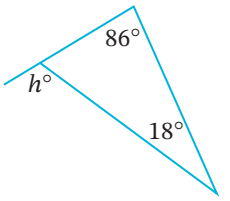
6.07

Foundation Standard Complex

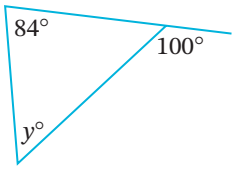
EXAMPLE 8

2 Find the value of each variable, giving reasons. R C

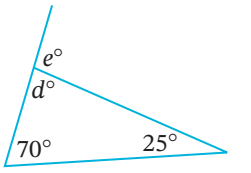
a



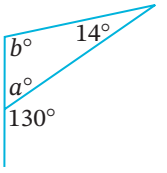
b



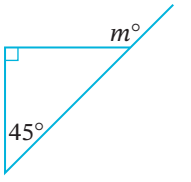
c



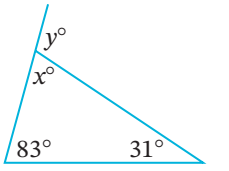
d



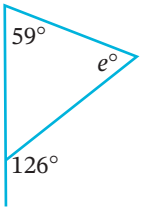
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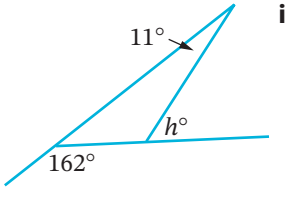
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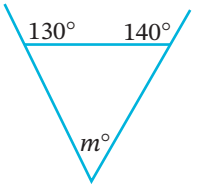
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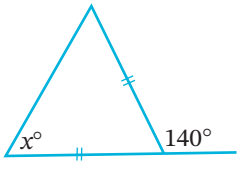
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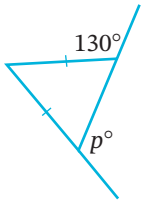
i



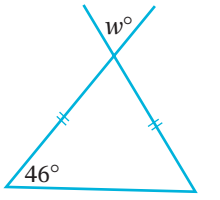
j



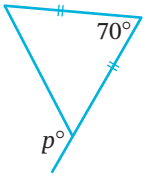
k



l



3 What is the value of p? Select the correct answer **A**, **B**, **C** or **D**.

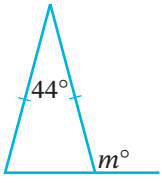


A 110 **B** 125

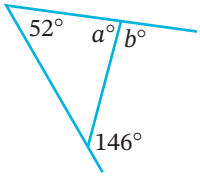
C 55 **D** 140

4 Find the value of each variable, giving reasons. R C

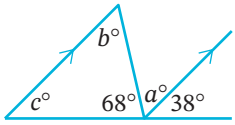
a



b



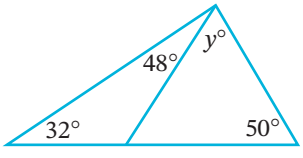
c



5 Find the value of y. Select **A**, **B**, **C** or **D**. R

A 80 **B** 50

C 130 **D** 230

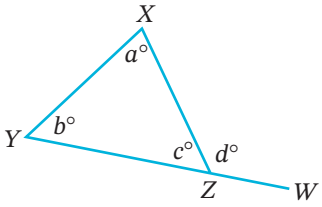


Foundation Standard Complex

9780170477963

6 Another exterior angle proof

Copy and complete the missing reasons in this proof.
Consider any triangle XYZ in which the angles are $a°$, $b°$ and $c°$. Extend the interval YZ to the point W. R C



$a° + b° + c° = 180°$ because _____.

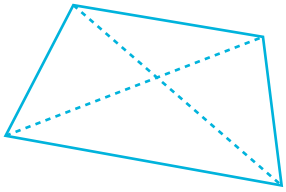
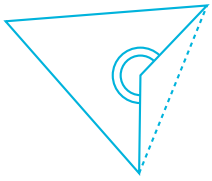
$\therefore a° + b° = 180° - c°$

But $d° = 180° - c°$ because _____.

$\therefore d° = a° + b°$

Classifying quadrilaterals 6.08

A **quadrilateral** is any shape with 4 straight sides.
A quadrilateral may be **convex** or non-convex (also called **concave**).

Convex quadrilateral	Non-convex quadrilateral
	
• All vertices (corners) point outwards. • All diagonals lie within the shape. • All angles are less than 180°.	• One vertex points inwards. • One diagonal lies outside the shape. • One angle is more than 180° (reflex angle).



Shutterstock.com/clivewa

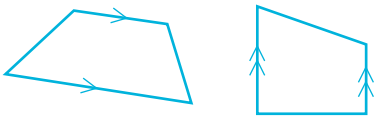
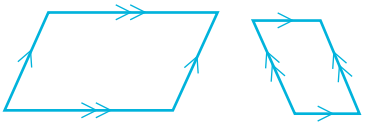
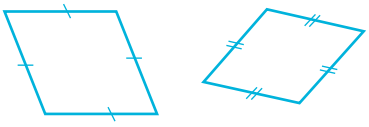
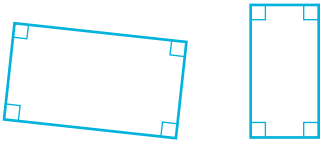
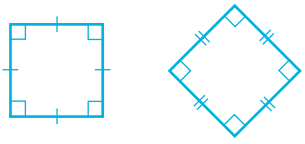
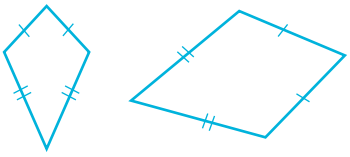
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There are 6 special types of quadrilaterals.

Worksheets
Classifying quadrilaterals
Constructing triangles and quadrilaterals

Puzzles
Constructions group clues
What shape am I?
Singing in the car

Skillsheet
Naming shapes

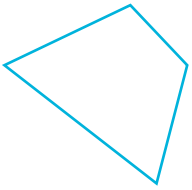
Name	A quadrilateral with:	Diagram
Trapezium	1 pair of opposite sides parallel	
Parallelogram	2 pairs of opposite sides parallel	
Rhombus (or diamond)	4 equal sides	
Rectangle	4 right angles	
Square	4 equal sides and 4 right angles	
Kite	2 pairs of adjacent sides equal	

EXERCISE 6.08 ANSWERS ON P. 581

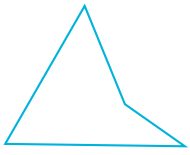
Classifying quadrilaterals U F R C

1 Is each quadrilateral convex or non-convex?

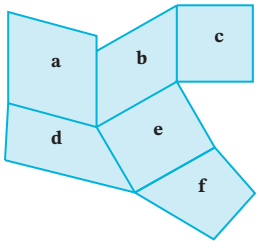
a 

b 

c 

d 

2 Name each quadrilateral in this diagram. **c**



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3 Find what the prefix 'quad' means. Find other words beginning with 'quad' and their meanings. C

4 Draw each of the special quadrilaterals and mark all axes of symmetry.

a rectangle **b** square **c** parallelogram
d rhombus **e** trapezium **f** kite

5 Copy and complete this table. R

Quadrilateral	Number of axes of symmetry	Order of rotational symmetry
	4	
Parallelogram		2
Trapezium		No rotational symmetry
Rhombus		2
	1	
	2	

6 Name all quadrilaterals that have: R C

a all angles equal to 90° **b** opposite sides parallel
c opposite sides equal **d** one pair of parallel sides
e 4 equal sides **f** 2 pairs of adjacent sides equal
g no axes of symmetry **h** rotational symmetry of order 2

7 Name the quadrilateral that has opposite sides equal and only 2 axes of symmetry. Select the correct answer **A**, **B**, **C** or **D**. R C

A rectangle **B** square **C** parallelogram **D** kite

Technology

Measuring quadrilaterals

In this activity, we use dynamic geometry software to construct quadrilaterals.

Square

1 Use the Regular polygon function to draw and label a square of length 7.5 cm.

2 Construct the 2 diagonals for the square. Measure the length of each diagonal. What do you notice?

3 Measure each angle of the square. What do you notice?

4 Draw another square with a side length of 9 cm. Repeat steps **2** and **3**. List 2 properties of the square.



Worksheet
Constructing triangles and quadrilaterals

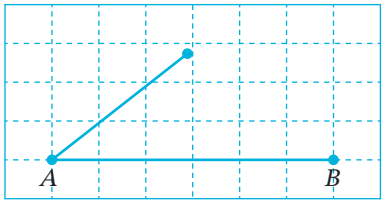
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Rectangle

1. Use the Polygon function to draw and label a rectangle with sides 6 cm by 3 cm.
2. Measure the sides and angles of the rectangle to check your accuracy.
3. Draw another rectangle that is 5 cm by 8.4 cm. Measure the sides and angles.
4. Measure the lengths of the 2 diagonals for each rectangle. What do you notice?
5. Copy and complete this property: The _____ in a rectangle are _____.
6. For the diagonals of the first rectangle, measure the distances to the point where they cross. Do they bisect each other or not? Repeat for the second rectangle.
7. Copy and complete: The _____ of a rectangle _____ each other.

Parallelogram

1. Draw and label an interval *AB* 6 cm long. Label the vertices.
2. From point A, draw an interval of 4 cm. Label the new point, C.



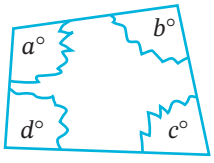
3. Draw a line parallel to AB through point C.
4. Create the 4th vertex of the parallelogram. Label the vertex D.
5. Measure the length of each side of the parallelogram. What do you notice?
6. Copy and complete: The _____ sides of a parallelogram are _____.
7. Measure the size of ∠CAB and ∠CDB. Repeat for ∠ACD and ∠ABD. What do you notice?
8. Copy and complete: The _____ angles of a parallelogram are _____.
9. Draw and measure the lengths of the diagonals of the parallelogram. Are they equal?
10. Copy and complete: The diagonals of a parallelogram are _____.
11. Do the diagonals of a parallelogram bisect each other or not?
12. Copy and complete: The diagonals of a parallelogram _____ each other.
13. Drag any vertices of the parallelogram that you can. Is it possible to draw other parallelograms with the same dimensions, 6 cm by 4 cm? What do you notice?
14. Now accurately construct other quadrilaterals such as a rhombus, kite or trapezium.

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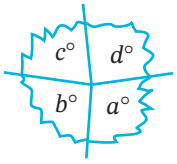
Investigation

Angle sum of a quadrilateral

What is the sum of the 4 angles in a quadrilateral? In groups of 2 to 4, complete the following activity.



1. Draw any large quadrilateral on paper, cut it out and label its 4 angles a, b, c or d.
2. Tear off the 4 angles and arrange them so that their points meet.
3. What type of angle do they form? How many degrees are in this angle?
4. To see if this works for all quadrilaterals, repeat the above steps for different quadrilaterals.



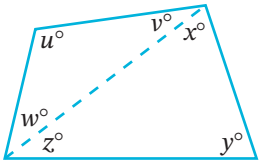
Angle sum of a quadrilateral 6.09

Any quadrilateral can be divided into 2 triangles along one of its **diagonals**. Because the angles in each triangle add to 180°, the angles in both triangles add to $2 \times 180° = 360°$.

$u° + v° + w° = 180°$ and $x° + y° + z° = 180°$

$\therefore$ Angle sum of a quadrilateral $= 180° + 180°$

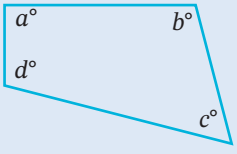
$= 360°$



Angle sum of a quadrilateral

The angle sum of a quadrilateral is 360°.

$$a + b + c + d = 360$$



Puzzles
Find the unknown angle

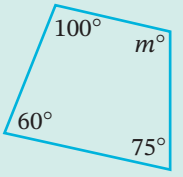
Mixed angle problems

Triangles and quadrilaterals

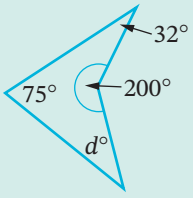
Example 9

Find the value of each variable, giving reasons.

a



b



Solution

a $m + 75 + 60 + 100 = 360$ (angle sum of a quadrilateral)

$m + 235 = 360$

$m = 360 - 235$

$= 125$

b $d + 75 + 32 + 200 = 360$ (angle sum of a quadrilateral)

$d + 307 = 360$

$d = 360 - 307$

$= 53$

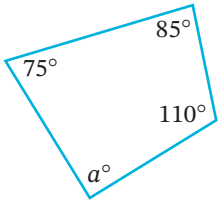
EXAMPLE 9

EXERCISE 6.09 ANSWERS ON P. 581

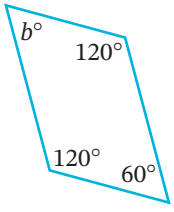
Angle sum of a quadrilateral U F PS R C

1 Find the value of each variable. R C

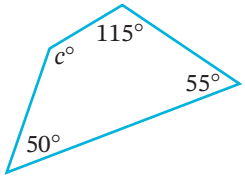
a



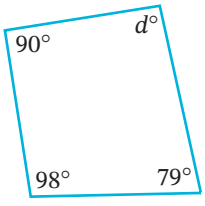
b



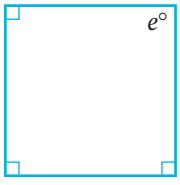
c



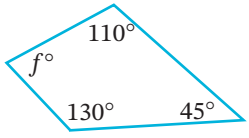
d



e



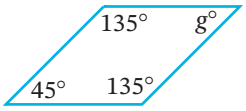
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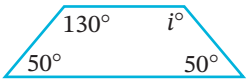
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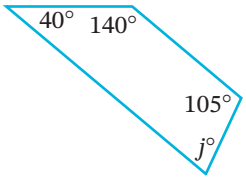
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h



i



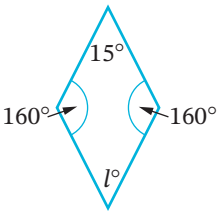
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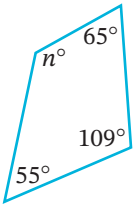
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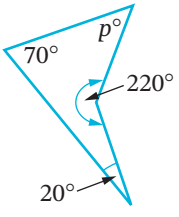
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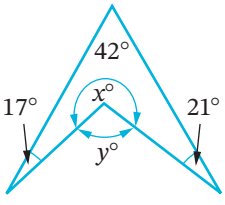
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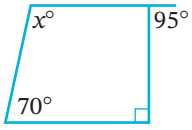
n



o



2 What is the value of x? Select the correct answer **A**, **B**, **C** or **D**. R



A 85 **B** 95

C 105 **D** 115

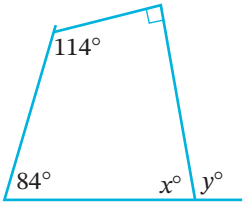
3 Sketch a rhombus. If one of the angles is 75°, what are the sizes of the other 3 angles? Select **A**, **B**, **C** or **D**. R

A 75°, 105°, 105° **B** 95°, 95°, 95°

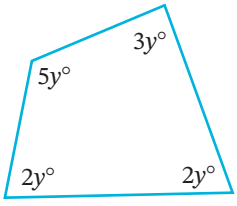
C 75°, 75°, 75° **D** 75°, 15°, 15°

4 Find the value of each variable, giving reasons. PS R C

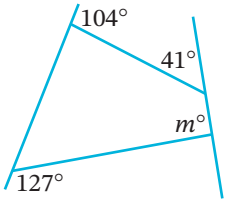
a



b



c



5 Robert's favourite quadrilateral is equiangular, meaning that all of its angles are equal. R C

a What is the size of each angle?

b What is the most general name for Robert's favourite quadrilateral?

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6.10 Properties of quadrilaterals

Worksheets
Properties of quadrilaterals

Diagonal properties of quadrilaterals

Always, sometimes, never true?

Puzzle
Geometry crossword

Videos
Classifying quadrilaterals

Properties of quadrilaterals

EXERCISE 6.10 ANSWERS ON P. 581

Properties of quadrilaterals U F R C

Questions **1** to **5** can also be found on the worksheet *Properties of quadrilaterals.*

1 Draw and cut out a trapezium, parallelogram, rhombus, rectangle, square and kite or use the ones supplied on the worksheet.

2 **a** By folding and measuring with a ruler, protractor and set square, discover the properties of each quadrilateral from question **1**. Copy the table or use the one supplied on the worksheet.

	Trapezium	Parallelogram	Rhombus	Rectangle	Square	Kite
Opposite sides are equal						
Opposite sides are parallel						
Opposite angles are equal						
All angles are 90°						
Diagonals are equal						
Number of axes of symmetry						
Order of rotational symmetry						

Complete the table by placing a tick in the appropriate place if the property is true for that quadrilateral and writing a number for the 'Number of axes of symmetry' and 'Order of rotational symmetry' rows.

b Ask your teacher to check your results.

c You will notice that there were no ticks for the kite. Write 2 properties of the kite (that is, 2 things that are always true about its sides, angles or diagonals). R C

3 Draw each quadrilateral and mark all axes of symmetry,

4 List the quadrilaterals that have rotational symmetry and mark the centre of symmetry, *O*, for each one.

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5 **a** Draw the diagonals of each quadrilateral from question **1**.

b By measuring with a ruler, protractor and set square, discover the properties of the diagonals of each quadrilateral as listed in the table below. Copy this table or use the one on the worksheet. Complete the table by placing ticks for each shape.

	Trapezium	Parallelogram	Rhombus	Rectangle	Square	Kite
Diagonals are equal						
Diagonals bisect each other						
Diagonals intersect at right angles						
Diagonals bisect angles of quadrilateral						

c Ask your teacher to check your results. R C

6 I am a quadrilateral with equal and parallel opposite sides. My diagonals are equal and I have 4 axes of symmetry. Which quadrilateral am I? Select **A**, **B**, **C** or **D**. R C

A rectangle **B** square **C** parallelogram **D** rhombus

7 Which quadrilateral am I? (There may be more than one answer.) R C

a My diagonals are equal.

b My diagonals bisect each other.

c I have 4 right angles.

d I have 2 pairs of parallel opposite sides.

e I have rotational symmetry, but no axes of symmetry.

f My diagonals cross each other at right angles.

g My diagonals bisect each other at right angles

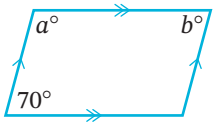
h My diagonals are equal and bisect each other.

8 Copy and complete the blanks to find the values of a and b. R C

$a + 70 = 180$ (_____ angles on _____ lines)

$a =$ _____

$b =$ _____ (opposite _____ of a parallelogram)

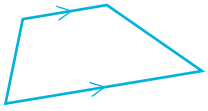
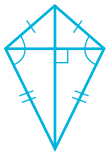
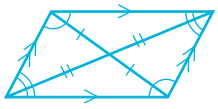
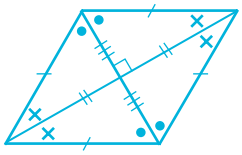
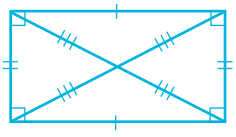
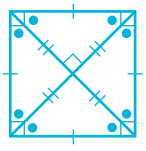


9 **a** Does a square have all the properties of a rectangle? Why?

b Does a rhombus have all the properties of a parallelogram? Why? R C

Foundation Standard Complex

10 Copy and complete this table of properties of quadrilaterals.

Shape	Properties
Trapezium 	• One pair of _____ sides • No axes of symmetry
Kite 	• _____ pairs of adjacent sides are equal • One pair of opposite angles are _____ • Has _____ axis of symmetry • Diagonals intersect at _____ angles
Parallelogram 	• _____ sides are equal and parallel • Opposite angles are _____ • No axes of symmetry • Diagonals _____ each other
Rhombus 	• All 4 sides are_____ • _____ sides are parallel • Opposite angles are _____ • Has _____ axes of symmetry • _____ bisect at right angles • Diagonals bisect the _____ of the rhombus
Rectangle 	• All 4 angles measure _____ • Opposite sides are _____ and _____ • Has 2 axes of _____ • Diagonals are _____ • _____ bisect each other
Square 	• All 4 sides are _____ • All 4 angles measure _____ • Has _____ axes of symmetry • Diagonals are equal and _____ each other at right angles • _____ bisect the angles of the square

Foundation Standard Complex

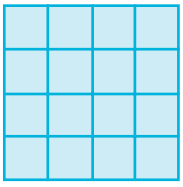
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Investigation

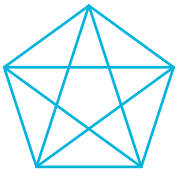
Shape puzzles

1 **a** How many squares can you find in this diagram?
The answer is not 16!

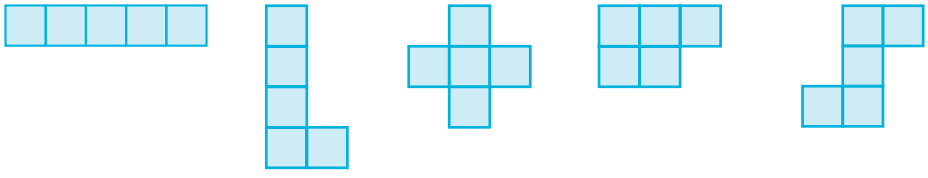
b How many rectangles can you find?



2 Can you trace this shape without going over any line twice and without lifting your pencil from the paper?

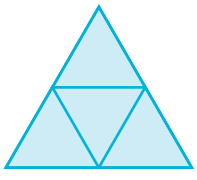


3 There are 12 different ways to join 5 squares edge to edge. These shapes are called pentominoes. Here are 5 of them. Draw the other 7.

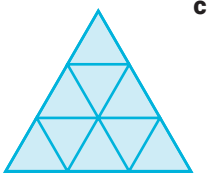


4 How many triangles can you find in each diagram?

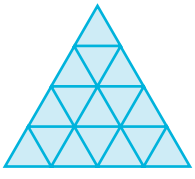
a



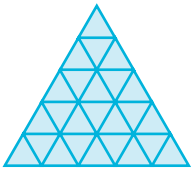
b

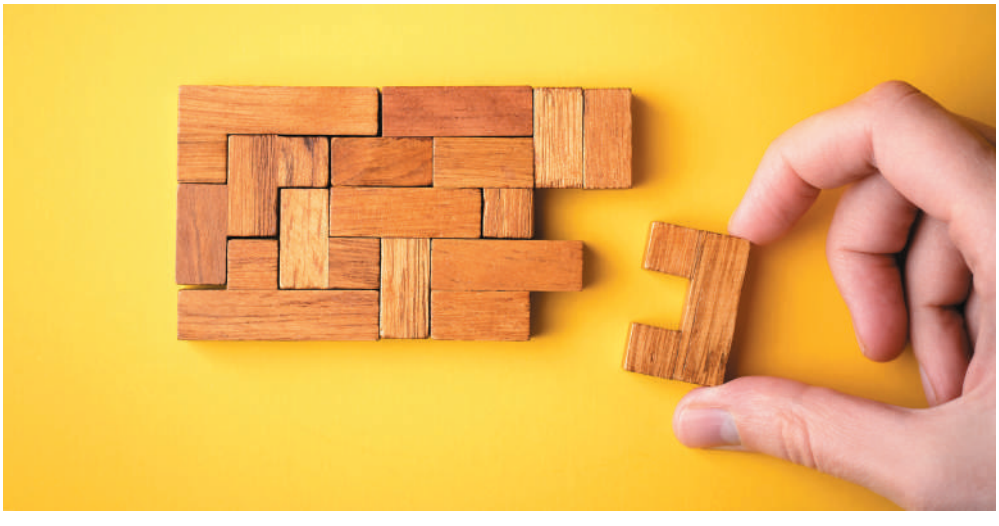


c



d





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Puzzle
Shapes puzzle 1

6.10

Worksheet
Equal angles

Power plus ANSWERS ON P. 582

1 Use the definitions of the quadrilaterals in the table on page 261 to decide whether:
   - **a** a square is a special type of rhombus
   - **b** a rhombus is a special type of square
   - **c** a parallelogram is a special type of trapezium
   - **d** a rectangle is a special type of parallelogram
   - **e** a parallelogram is a special type of kite
   - **f** a rectangle is a special type of square

2 **a** What additional property makes a parallelogram a rectangle?
   - **b** What makes a kite a rhombus?
   - **c** What makes a rectangle a square?

3 Name the most general quadrilateral in which:
   - **a** opposite angles are equal
   - **b** diagonals intersect at 90°
   - **c** diagonals are equal
   - **d** all angles are 90°
   - **e** opposite sides are parallel
   - **f** diagonals bisect each other.

4 Investigate the angle sum of a pentagon.

5 **a** Draw any quadrilateral and extend each side to create 4 exterior angles.
   - **b** Investigate the sum of the exterior angles.

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CHAPTER 6 REVIEW

Quiz
Language of maths 6

Puzzle
Geometry find-a-word

Language of maths

acute-angled	angle sum	axis/axes of symmetry	convex
equilateral	exterior angle	image	isosceles
kite	line symmetry	obtuse-angled	parallelogram
quadrilateral	rectangle	reflection	rhombus
right-angled	rotation	rotational symmetry	scalene
square	transformation	translation	trapezium

1 What are the 2 types of symmetry?

2 When you look into a mirror, you see your reflection. Is this meaning of 'reflection' related to its meaning in this chapter?

3 A **regular** shape has equal sides and equal angles. What is the more common name for a **regular quadrilateral**?

4 The word 'isosceles' comes from Greece. Use a dictionary to find out what it means in Greek.

5 Use a dictionary to find the different meanings of:

a translation **b** axis

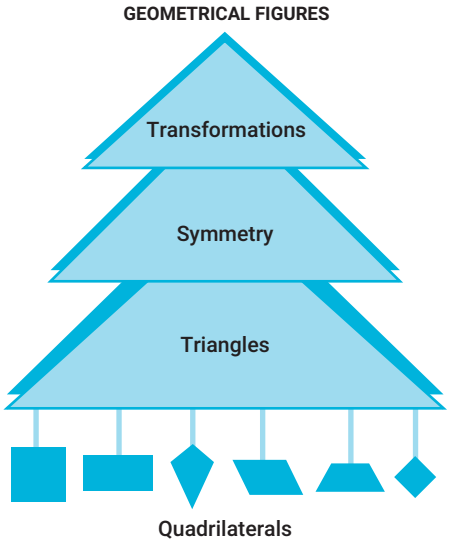
6 Name the shape that has 2 pairs of equal adjacent sides.

Topic summary

Worksheet
Mind map: Geometrical figures

- How useful do you think this chapter will be to you in the future?
- Can you name any jobs that use some of the concepts covered in this chapter?
- Did you have any problems with any sections of this chapter? Discuss any problems with a friend or your teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Quiz
Test yourself 6

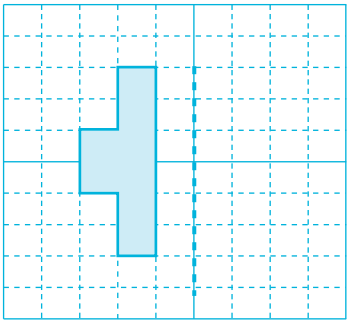
TEST YOURSELF 6

ANSWERS ON P. 582

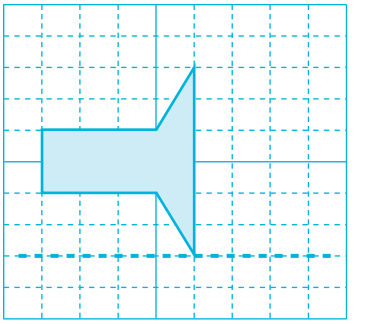
6.01

1 Copy and reflect each figure about the dotted line.

a



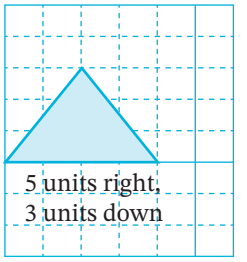
b



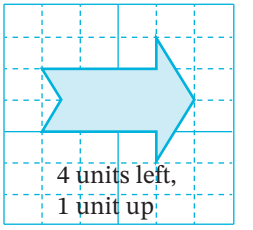
6.01

2 Copy and translate each figure by the given distance and direction.

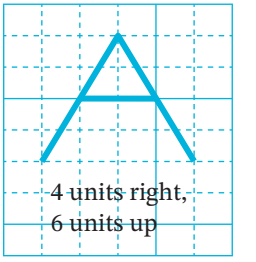
a



b



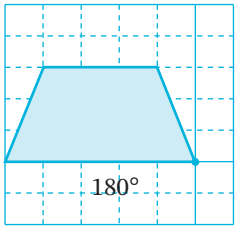
c



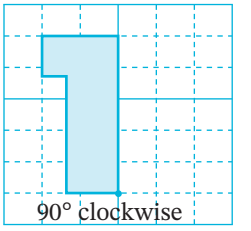
6.01

3 Copy and rotate each figure about the given point by the stated angle.

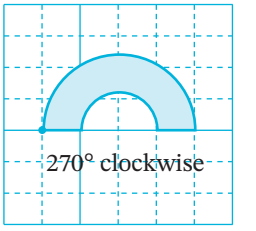
a



b

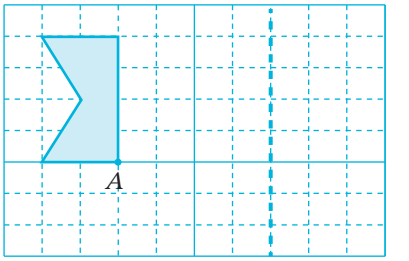


c



6.02

4 Translate the shape 5 units to the right and 2 units up.
Then rotate it 90° clockwise about vertex A and reflect it across the given dotted line.



Foundation | Standard | Complex

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5 Copy each figure and draw its axes of symmetry. 6.03

a



b



c

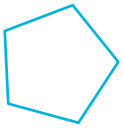


6 Test whether each shape has rotational symmetry. If it does, write the order of rotational symmetry. 6.04

a



b

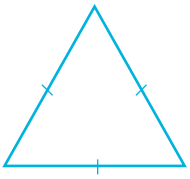


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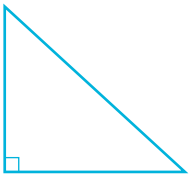


7 Classify each triangle by sides and by angles. 6.05

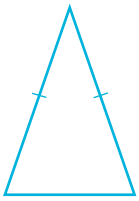
a



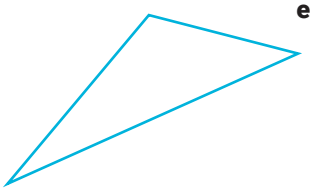
b



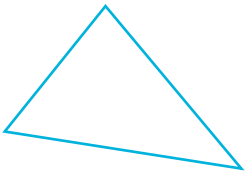
c



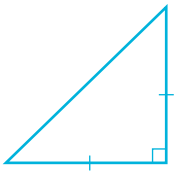
d



e

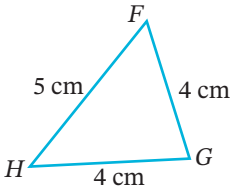


f



8 **a** Classify $\triangle FGH$ by sides and by angles. 6.05

b Which angles in $\triangle FGH$ are equal?



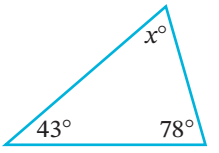
9 Classify each described triangle by sides and by angles.

a All of my angles are equal.

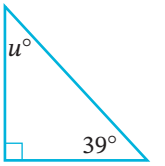
b My angles are 60°, 80° and 40°.

6.06 **10** Find the value of each variable, giving reasons.

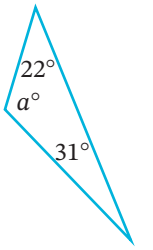
a



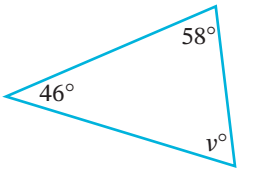
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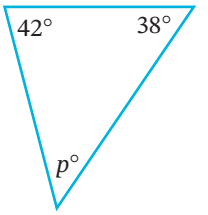
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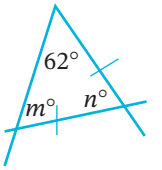
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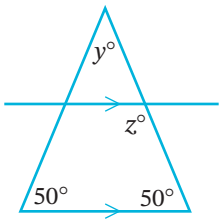
e



f



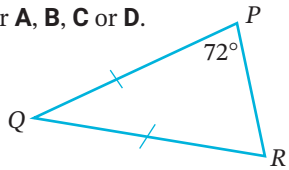
g



6.06 **11** Find the size of $\angle PQR$. Select the correct answer **A**, **B**, **C** or **D**.

A 18° **B** 36°

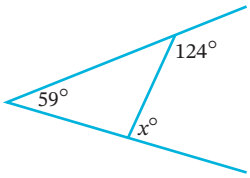
C 54° **D** 72°



6.07 **12** Find x. Select **A**, **B**, **C** or **D**.

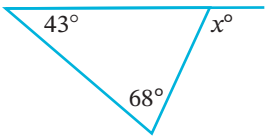
A 59 **B** 65

C 115 **D** 124

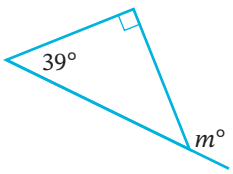


6.07 **13** Find the value of each variable, giving reasons.

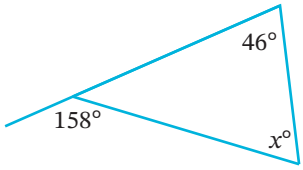
a



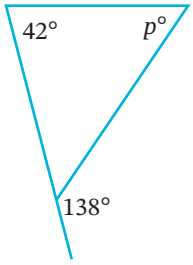
b



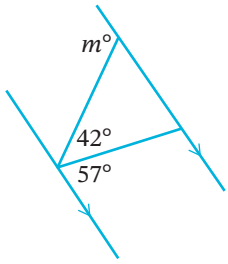
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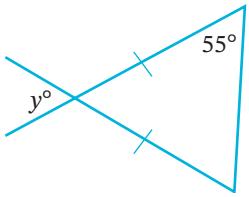
d



e



f

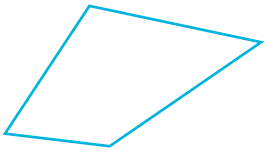


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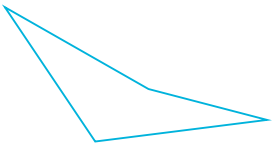
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14 Is each quadrilateral convex or non-convex? 6.08

a

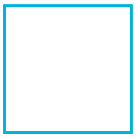


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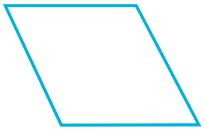


15 Name each special quadrilateral. 6.08

a



b



c



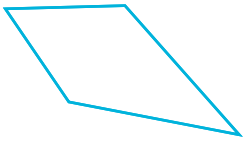
d



e



f



16 **a** Copy each quadrilateral in question **15** and mark all the axes of symmetry. 6.08

b State which quadrilaterals have rotational symmetry, and their order of rotational symmetry.

17 What is the definition of a rhombus? 6.08

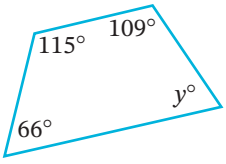
18 What quadrilateral am I? (Answer with the most general quadrilateral). 6.08

a I have opposite sides parallel.

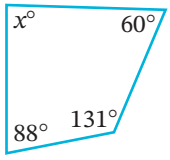
b I have one pair of parallel sides.

19 Find the value of each variable, giving reasons. 6.09

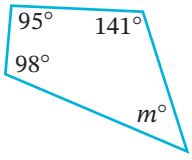
a



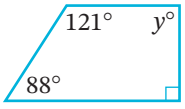
b



c



d



20 *EFGH* is a parallelogram. Which statement is NOT true? Select **A**, **B**, **C** or **D**. 6.10



A $\angle EFG = \angle EHG$ **B** $EF = HG$

C $EF = FG$ **D** $\angle EFG + \angle FGH = 180°$

21 List 2 properties of a parallelogram. 6.10

22 What quadrilateral am I? (Answer with all possible quadrilaterals).

a All of my angles are equal.

b My diagonals are equal and bisect each other.

c My diagonals bisect each other.

d My opposite angles are equal.

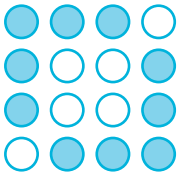
Foundation | Standard | Complex

PRACTICE SET 2

ANSWERS ON P. 583

4.01

1 What fraction of these circles are white?
Select the correct answer **A**, **B**, **C** or **D**.



A $\frac{1}{3}$ **B** $\frac{1}{2}$ **C** $\frac{3}{8}$ **D** $\frac{9}{16}$

4.03

2 Evaluate each expression.

a $\frac{1}{2}+\frac{1}{3}$ **b** $\frac{1}{3}+\frac{2}{5}$ **c** $\frac{1}{2}-\frac{3}{8}$

4.10

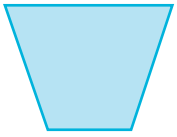
3 Convert each fraction to a percentage.

a $\frac{1}{4}$ **b** $\frac{3}{5}$ **c** $\frac{23}{40}$

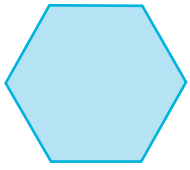
6.03

4 Copy each shape and mark in its axes of symmetry.

a



b



c



5.03

5 Simplify each expression.

a $w \times 2$ **b** $a \times a$ **c** $4 \times a \times b \times k$ **d** $p + p + p + p$

e $m \div 2$ **f** $h - h$ **g** $6k + 2k$ **h** $5 \times m \times 2$

4.01

6 Change each improper fraction to a mixed numeral.

a $\frac{12}{5}$ **b** $\frac{7}{2}$ **c** $\frac{5}{3}$ **d** $\frac{11}{6}$

4.01

7 Simplify each fraction.

a $\frac{8}{10}$ **b** $\frac{12}{18}$ **c** $\frac{15}{24}$ **d** $\frac{70}{100}$

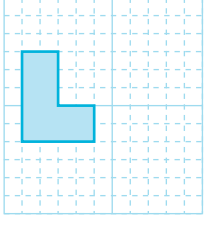
5.01

8 Evaluate each expression mentally.

a $32 + 24 + 6$ **b** 2×900 **c** $-12 + 22$ **d** $31 \times 5 \times 4$

6.01

9 Copy this L-shape on grid paper and translate it 5 units right and 2 units down.

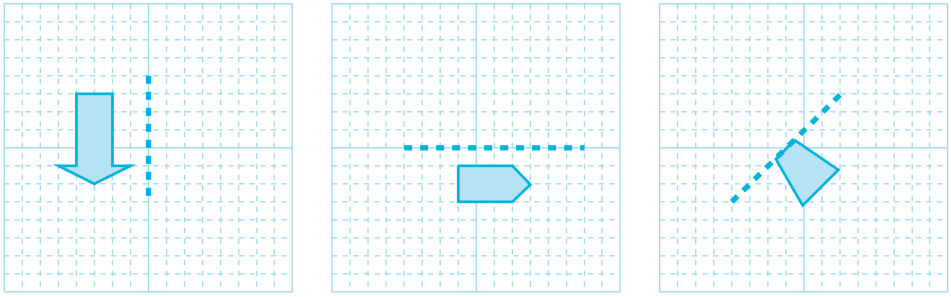


Worksheet
2 mm grid paper

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10 Copy each shape on grid paper and reflect it across the line.



6.01

Worksheet
2 mm grid paper

11 Copy and complete each statement with a > or < symbol.

a $\frac{1}{3} ____ \frac{1}{4}$ **b** $\frac{2}{5} ____ \frac{5}{10}$ **c** $\frac{5}{7} ____ \frac{3}{5}$

4.02

12 Evaluate each product without using a calculator.

a 18×11 **b** 24×8 **c** 15×9 **d** 22×12

5.02

13 Evaluate each expression.

a $5\frac{3}{4} + 1\frac{2}{3}$ **b** $4\frac{3}{5} - 1\frac{1}{2}$ **c** $1\frac{2}{3} - \frac{7}{8}$

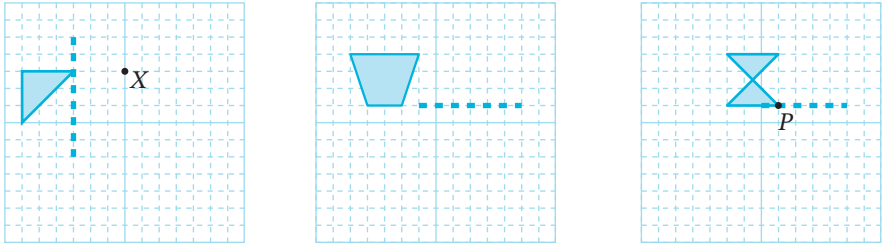
4.04

14 Copy each shape on grid paper and draw the image after the composite transformations.

a Reflect across the line, rotate 90° clockwise about *X*.

b Translate 5 units right, then reflect across the line.

c Rotate 90° clockwise about *P*, then reflect across the line, then translate 3 units left.



6.02

Worksheet
2 mm grid paper

15 Evaluate each product.

a $\frac{3}{5} \times \frac{1}{2}$ **b** $\frac{2}{3} \times \frac{7}{12}$ **c** $3\frac{1}{3} \times 2\frac{1}{2}$

4.06

16 Convert each decimal to a percentage.

a 0.4 **b** 0.05 **c** 0.12

4.10

17 Find:

a 5% of $200

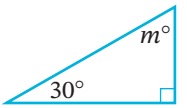
b 30% of 1 hour (in minutes)

4.11

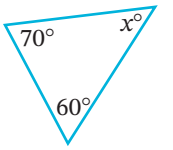
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6.06 **18** Find the value of the variable in each triangle.

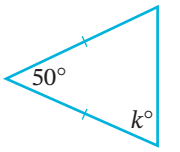
a



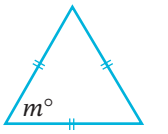
b



c



d



6.05 **19** Classify each triangle in question **18** above by sides and by angles.

5.05 **20** Evaluate each expression.

a $m - 3$ if $m = -2$ **b** $2p - 1$ if $p = 10$ **c** $\frac{2d}{3}$ if $d = 12$

5.11 **21** Solve each equation.

a $x + 11 = 34$ **b** $3x = -27$ **c** $d - 3 = -9$ **d** $\frac{m}{5} = 1$

4.05 **22** Find $\frac{3}{8}$ of 2 m (in cm).

4.12 **23** Chloe owns 40 pairs of shoes. 2 pairs are black.

a What fraction of Chloe's shoes are black?

b What percentage of Chloe's shoes are black?

5.11 **24** Solve the equation $108x = 18$. Select **A**, **B**, **C** or **D**.

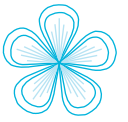
A $x = 21$ **B** $x = 15$ **C** $x = 6$ **D** $x = \frac{1}{6}$

6.04 **25** For each shape:

i count how many axes of symmetry it has

ii state whether it has rotational symmetry and if it does, state the order.

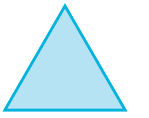
a



b



c



4.07 **26** Evaluate each quotient.

a $\frac{3}{4} \div \frac{1}{2}$ **b** $\frac{2}{10} \div 5$ **c** $2\frac{1}{2} \div \frac{3}{5}$

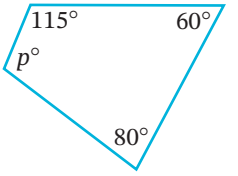
5.04 **27** Write an algebraic expression for:

a the product of p and q **b** 4 less than x **c** d decreased by 7

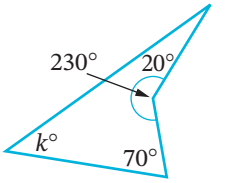
d the number of girls if there are b boys in a class of 28 students

6.09 **28** Find the value of each variable.

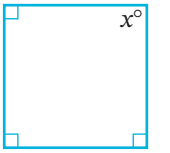
a



b



c



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29 Simplify each expression. 5.07

a $8k + 2j - k + 9j$ **b** $10p - p$

c $14p - 3q - 7p + 2q$ **d** $5x + 12y - x - 4y$

30 Simplify each expression. 5.08-5.09

a $7 \times 2p$ **b** $12m \div 3m$ **c** $-5x \times 5xy$

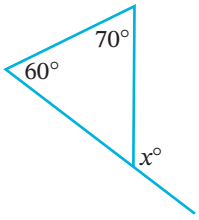
d $3a \times (-b) \times 4$ **e** $\frac{4x^2y}{24xy}$ **f** $\frac{5a \times 9a}{15a}$

31 Solve each equation. 5.12

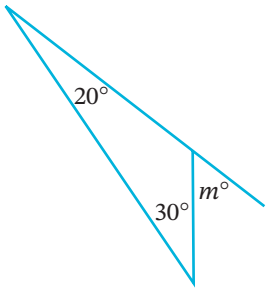
a $5m - 3 = 22$ **b** $6n + 3 = 12$ **c** $-3x - 11 = -20$ **d** $\frac{x}{2} + 5 = 9$

32 Find the value of each variable. 6.07

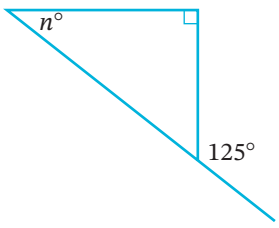
a



b



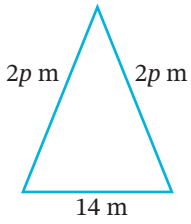
c



33 Use an equation to solve each problem. 5.13

a If you multiply Mr Ha's age by 4 and add 16, you obtain 200. How old is Mr Ha? (Let h be his age.)

b The perimeter of this isosceles triangle is 56 cm. Find the value of p.



34 Name all quadrilaterals (parallelogram, trapezium, rectangle, kite, rhombus, square) that have: 6.08

a opposite sides parallel

b 4 right angles

c 4 equal sides and 4 equal angles

d 2 pairs of equal adjacent sides

35 Which number could replace the △ in $\frac{\triangle+3}{5}$ to make the fraction between 6 and 7? Select **A**, **B**, **C** or **D**. 4.01

A 26 **B** 32 **C** 31 **D** 36

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